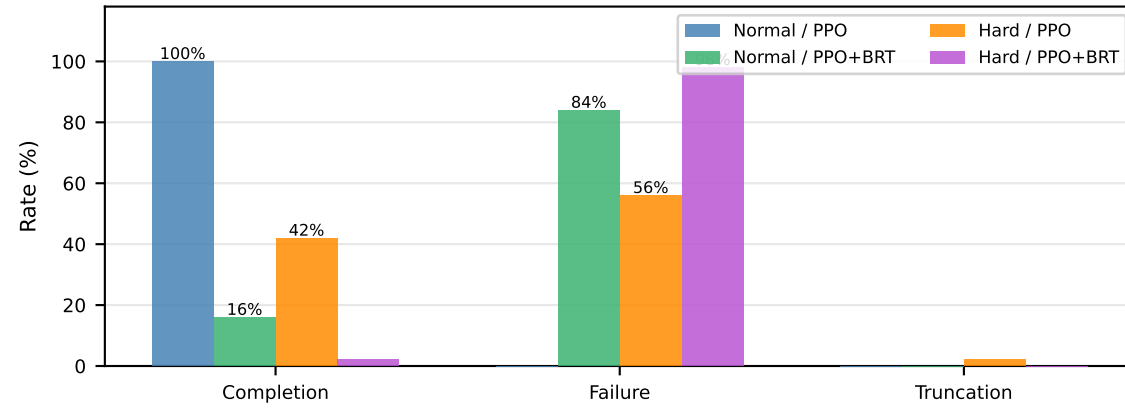
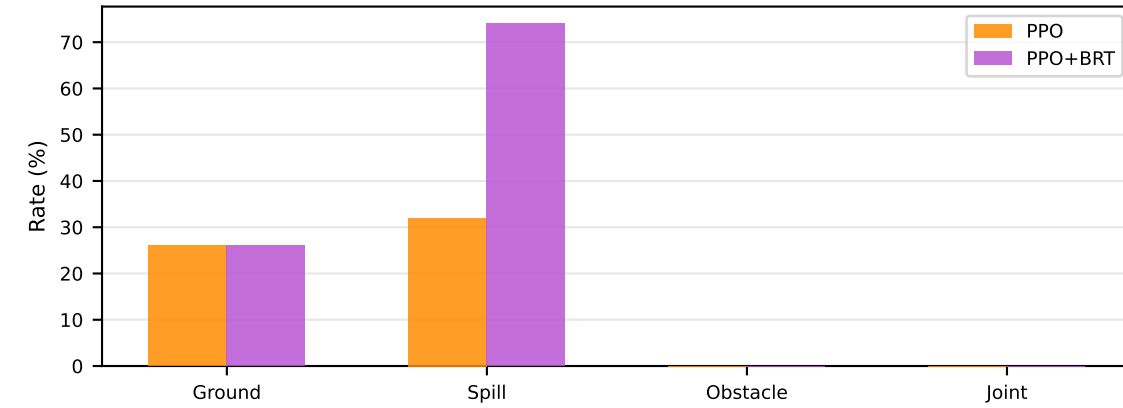


**Safety evaluation: PPO vs PPO + BRT safety filter**  
**PPO hard-start failure 56% | PPO+BRT hard-start failure 98%**

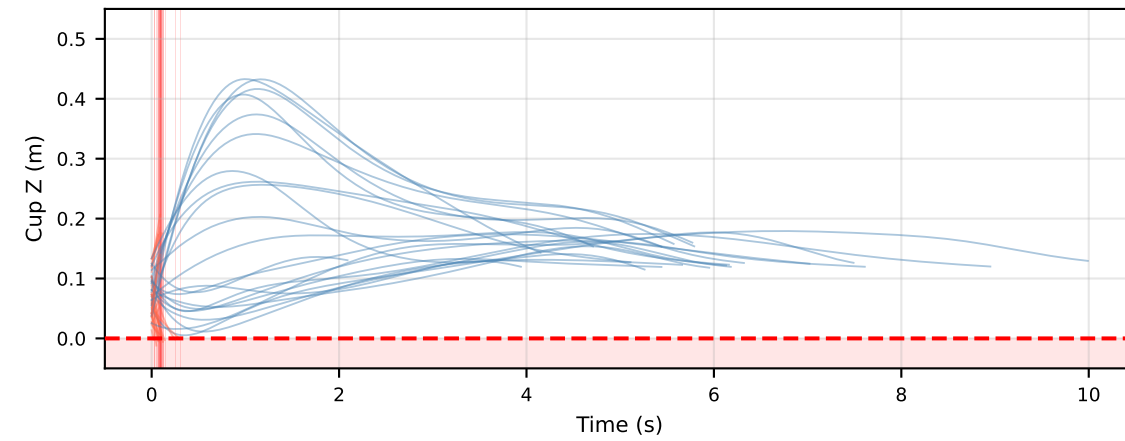
**(A) Episode outcomes**



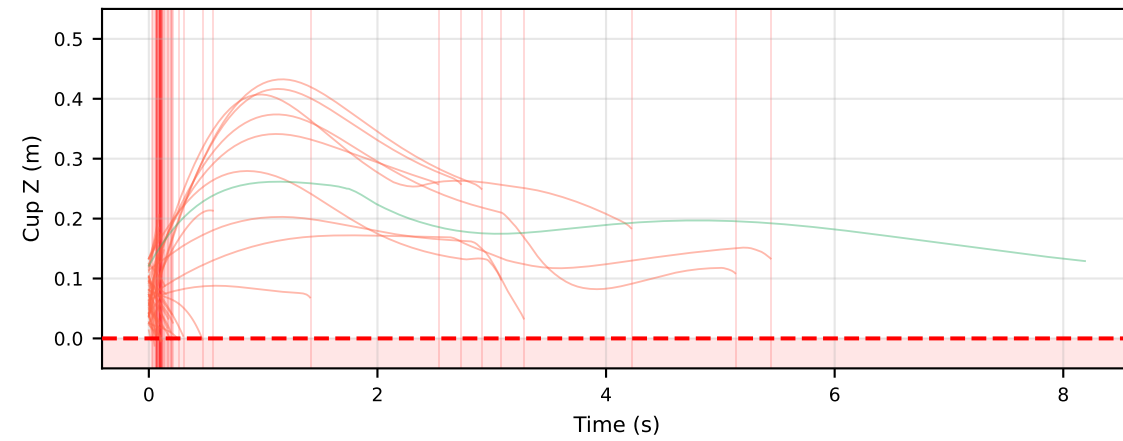
**(B) Failure type breakdown (hard starts)**



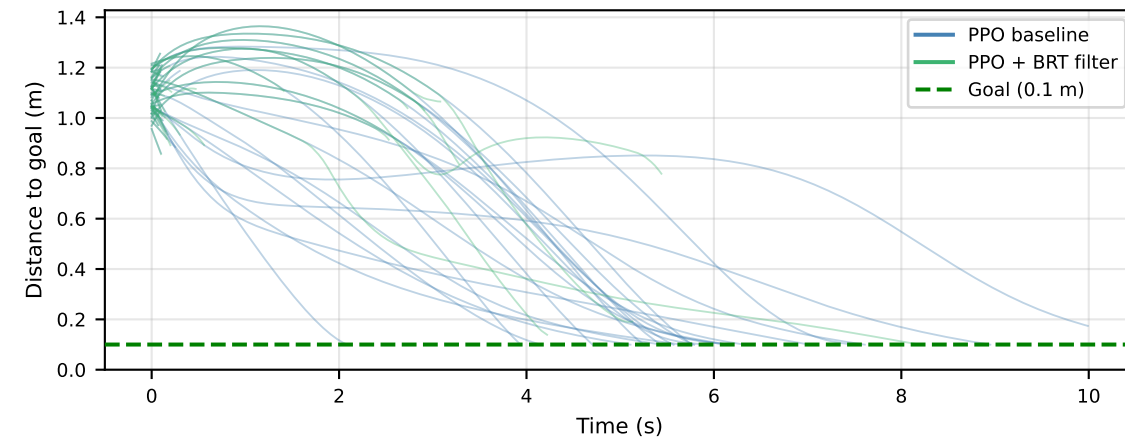
**(C) Hard starts — PPO baseline (cup height)**



**(D) Hard starts — PPO + BRT filter (cup height)**



**(E) Progress toward goal (hard starts)**



**(F) BRT filter activity (hard starts)**

